

RAG for Financial Question Answering: A Comparative Evaluation Against Standalone LLMs on SEC 10-K Filings

Design and Specification Proposal
COMP702 — M.Sc. project (2024/25)

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1 Statement of Ethical Compliance: A0

Data Category: A

Participant Category: 0

This project does not involve human participants and uses only publicly available data — SEC 10-K annual filings from SEC EDGAR. The data category is A0 (no personal, sensitive, or confidential data). I confirm I will follow the University’s ethical guidance throughout.

2 Project Description

Large Language Models (LLMs) such as GPT-4 demonstrate strong general-purpose question answering, but struggle with fact-intensive tasks like financial analysis. When queried on exact revenue figures or specific risk disclosures, LLMs frequently produce plausible but incorrect answers, since such precise information is rarely memorised reliably during training.

This project investigates whether Retrieval-Augmented Generation (RAG) can address this. RAG combines an LLM with a searchable external knowledge base, grounding answers in actual source documents. The knowledge base consists of SEC 10-K filings for FY2023 from five companies: Apple, Microsoft, JPMorgan Chase, ExxonMobil, and Johnson & Johnson — spanning four industries (technology, banking, energy, and healthcare).

The system splits each 10-K into text chunks, converts them into a vector database, and retrieves relevant chunks when a question is asked. Those chunks are passed to the LLM as context, so answers are grounded in the filing text rather than the model’s internal knowledge. RAG answers are then compared against a standalone LLM — the same model without retrieved context — across manually written financial questions covering factual, numerical, and risk-based types. Performance is measured using RAGAS alongside Exact Match and F1 score.

While RAG is established, existing datasets such as FinQA use older financial reports (1999–2019). This project focuses on FY2023 10-K filings with a purpose-built evaluation dataset, providing a clearer picture of where retrieval augmentation helps and where it falls short — particularly with numerical reasoning and tabular data.

3 Aims and Requirements

3.1 Aims

- Design and implement a RAG pipeline for answering financial questions grounded in SEC 10-K filings.
- Quantitatively evaluate whether RAG improves accuracy, faithfulness, and relevance compared to a standalone LLM baseline.
- Analyse how RAG performance varies across factual, numerical, and risk-based question types, identifying where retrieval augmentation is most and least effective.

3.2 Requirements

Essential requirements:

- Download and parse SEC 10-K filings (HTML format) for five companies from SEC EDGAR.
- Clean and segment extracted text into overlapping chunks suitable for embedding.
- Embed text chunks and store them with metadata in a vector database.

- Retrieve the top- k most relevant chunks for a given question using semantic similarity search.
- Generate answers using an LLM conditioned on retrieved chunks (the RAG pipeline).
- Generate answers using the same LLM without retrieved context (the baseline pipeline) for direct comparison.
- Evaluate and compare RAG and baseline outputs using RAGAS (Faithfulness, Answer Relevance, Context Relevance) and Exact Match / F1 against a manually constructed evaluation dataset.

Desirable requirements:

- A simple interactive interface allowing a user to ask a question and view RAG and baseline answers side by side, including retrieved source passages.
- An ablation study comparing chunk sizes (256 vs 512 vs 1024 tokens) and their effect on retrieval quality.
- A breakdown of results by question type (factual, numerical, risk-based) and by company/industry.

4 Key Literature and Background Reading

Background reading spans four areas: RAG architecture, RAG evaluation, financial QA challenges, and long-context limitations.

4.1 Retrieval-Augmented Generation

Lewis et al. [1] introduced RAG, combining a pre-trained sequence-to-sequence model with a dense vector index. They showed RAG outperforms purely parametric models on open-domain QA — forming the architectural foundation of this project’s offline indexing / online retrieve-then-generate design.

Gao et al. [5] survey subsequent RAG development, distinguishing Naive, Advanced, and Modular RAG. This project uses Naive RAG as a starting point, with Advanced RAG techniques noted as future work. The survey also confirms that embedding models like OpenAI’s `text-embedding` family work best with 256–512 token chunks, shaping the chunking strategy used here.

4.2 Evaluating RAG Systems

Es et al. [2] introduced RAGAS, a reference-free evaluation framework scoring three dimensions: Faithfulness (is the answer grounded in retrieved context?), Answer Relevance (does the answer address the question?), and Context Relevance (is the retrieved context useful?). RAGAS showed strong agreement with human judgements on WikiEval (95% accuracy for Faithfulness) and is adopted here as the primary evaluation framework, supplemented by a manually annotated QA set and Exact Match / F1.

4.3 Financial Question Answering

Chen et al. [3] introduced FinQA, showing a large gap between model performance (61.2% execution accuracy) and human experts (91.2%), revealing the unique challenges of multi-step numerical reasoning over financial documents. These findings motivate this project’s design and inform the limitations analysis, since plain-text chunking handles tabular data poorly.

Daimi and Iqbal [6] proposed a framework for extracting and analysing 10-K filings using LLMs, confirming the feasibility of retrieving filings from SEC EDGAR and identifying comparable challenges in section identification and text extraction.

4.4 Long-Context Limitations

Liu et al. [4] showed that LLM performance degrades when relevant information sits in the middle of a long input — the “lost in the middle” effect. This shaped two decisions: using 512-token chunks to keep retrieved context compact, and setting $k=5$ to avoid passing large context volumes to the generator.

5 Preliminary Work

The project design and architecture have been finalised, and literature covering RAG, financial NLP, and evaluation methodology has been reviewed. The technology stack has been selected and justified. Preliminary investigation confirms that SEC EDGAR provides consistent HTML access to 10-K filings across the five target companies, and that Python’s `requests` and `BeautifulSoup` libraries are sufficient for retrieval and parsing. LangChain, Chroma, and OpenAI API access have been tested in a local environment to confirm compatibility.

6 Development and Implementation Summary

The project is implemented in Python using LangChain [10] to orchestrate the RAG pipeline. LangChain was chosen for its native integrations with Chroma, OpenAI models, and RAGAS, allowing focus on financial-domain design decisions rather than retrieval infrastructure. Development will be carried out in VS Code, with Git for version control and GitHub for backup.

Dense embedding-based retrieval is used rather than sparse keyword matching (e.g. BM25), following Karpukhin et al. [7], who showed dense retrieval outperforms sparse methods on knowledge-intensive QA tasks. Similarity search is implemented using approximate nearest-neighbour techniques [8].

Component	Tool / Model	Rationale
Filing retrieval & parsing	Python <code>requests</code> + <code>BeautifulSoup</code>	SEC EDGAR provides filings as HTML; BeautifulSoup extracts readable text without PDF overhead.
Chunking	LangChain <code>RecursiveCharacterTextSplitter</code> (512 tokens, 50 overlap)	Splits on paragraph/sentence boundaries; 512-token chunks align with optimal embedding performance [5] and reduce lost-in-the-middle risk [4].
Embedding model	OpenAI <code>text-embedding-3-small</code>	Strong retrieval performance at low cost; dense embeddings outperform sparse search on knowledge-intensive tasks.
Vector store	Chroma	Simple API, persists to disk, stores vectors, text, and metadata; sufficient at the scale of five filings.
Generator LLM	OpenAI GPT-4o-mini	Strong instruction-following at low cost; used identically for both RAG and baseline to ensure a fair comparison.
Evaluation	RAGAS + manual QA + Exact Match / F1	RAGAS provides automated reference-free scoring [2]; manual QA pairs with verified ground-truth answers enable direct EM/F1 comparison.

Table 1: Component summary and rationale

Development follows two stages: (1) an *offline indexing pipeline* that downloads, cleans, chunks, and embeds each filing once; and (2) an *online query pipeline* that embeds an incoming question, retrieves the top $k=5$ most similar chunks, and generates an answer. The same query is also sent to the LLM without retrieval to produce the baseline answer.

Code is version-controlled with Git, using dedicated branches for each pipeline stage (indexing, retrieval, evaluation) merged into `main` once tested. Development proceeds company by company — starting with Apple — so bugs can be caught before scaling to all five companies.

7 Data Sources

7.1 Knowledge Base Data

The RAG knowledge base consists of SEC 10-K filings for FY2023 from Apple Inc. (AAPL), Microsoft Corporation (MSFT), JPMorgan Chase & Co. (JPM), Exxon Mobil Corporation (XOM), and Johnson & Johnson (JNJ), retrieved from SEC EDGAR [9] — a free, publicly accessible system requiring only a descriptive User-Agent header. Five companies were chosen to span four distinct industries (technology, banking, energy, and healthcare), providing variety in financial terminology and reporting styles while keeping the dataset manageable within an MSc project timescale.

7.2 Evaluation Data

No existing benchmark covers these specific companies and the FY2023 period — the nearest comparable, FinQA [3], uses earnings reports from 1999–2019. A purpose-built evaluation dataset of approximately 50 question–answer pairs will be manually constructed from the five 10-K filings, covering factual, numerical, and risk-based question types (approximately 16–17 per category). This number enables meaningful comparison between the RAG and baseline systems while keeping manual ground-truth creation feasible within the project’s constraints, following the approach of Es et al. [2], who built their own WikiEval dataset to evaluate RAGAS. No personal, sensitive, or third-party data is involved.

8 Testing and Evaluation

8.1 Technical Testing

Each pipeline stage is tested incrementally. Data retrieval and cleaning scripts are validated on Apple’s filing first, with extracted text manually checked for completeness and absence of HTML artefacts before being applied to the remaining four companies. The chunking and embedding stage is verified by manually inspecting sample chunks and confirming retrieval returns expected source chunks for known test questions. Before full-scale evaluation, the end-to-end RAG pipeline is spot-checked to confirm that generated answers are grounded in retrieved context.

8.2 Evaluation of Project Aims

The core research question — whether RAG improves financial QA relative to a standalone LLM — is evaluated using the 50-question dataset described in Section 7. Both systems are run on every question and scored using:

- **RAGAS metrics:** Faithfulness and Answer Relevance for both systems; Context Relevance and Context Recall additionally for the RAG system.
- **Exact Match and F1 score** against verified ground-truth answers for both systems.

It should be noted that Exact Match is a conservative metric for this task: LLM-generated answers are full sentences rather than short extractive spans, so a correctly-phrased numerical answer such as “approximately \$383 billion” will score zero against a ground-truth of “\$383.3 billion” despite being substantively correct. F1 provides a more informative partial-credit signal for these longer answers. Accordingly, RAGAS scores (Faithfulness, Answer Relevance) are treated as the primary quality indicator, with EM/F1 as a secondary, stricter check.

Results are aggregated overall and broken down by question type (factual, numerical, risk-based) and by company. Where the two systems perform similarly on certain question types, this itself is a useful finding — suggesting parametric LLM knowledge is sufficient for those cases and retrieval adds little.

9 UI/UX Mockup

The core deliverable is the RAG pipeline and its evaluation rather than a polished application, but a simple comparison interface is planned as a desirable requirement to make results tangible. The wireframe shows a question input field, company selection chips, and two side-by-side result panels — the RAG answer (with retrieved source passages and RAGAS scores) and the standalone LLM answer (with its scores) — enabling direct visual comparison. The diagram also shows the underlying offline indexing and online query pipelines.

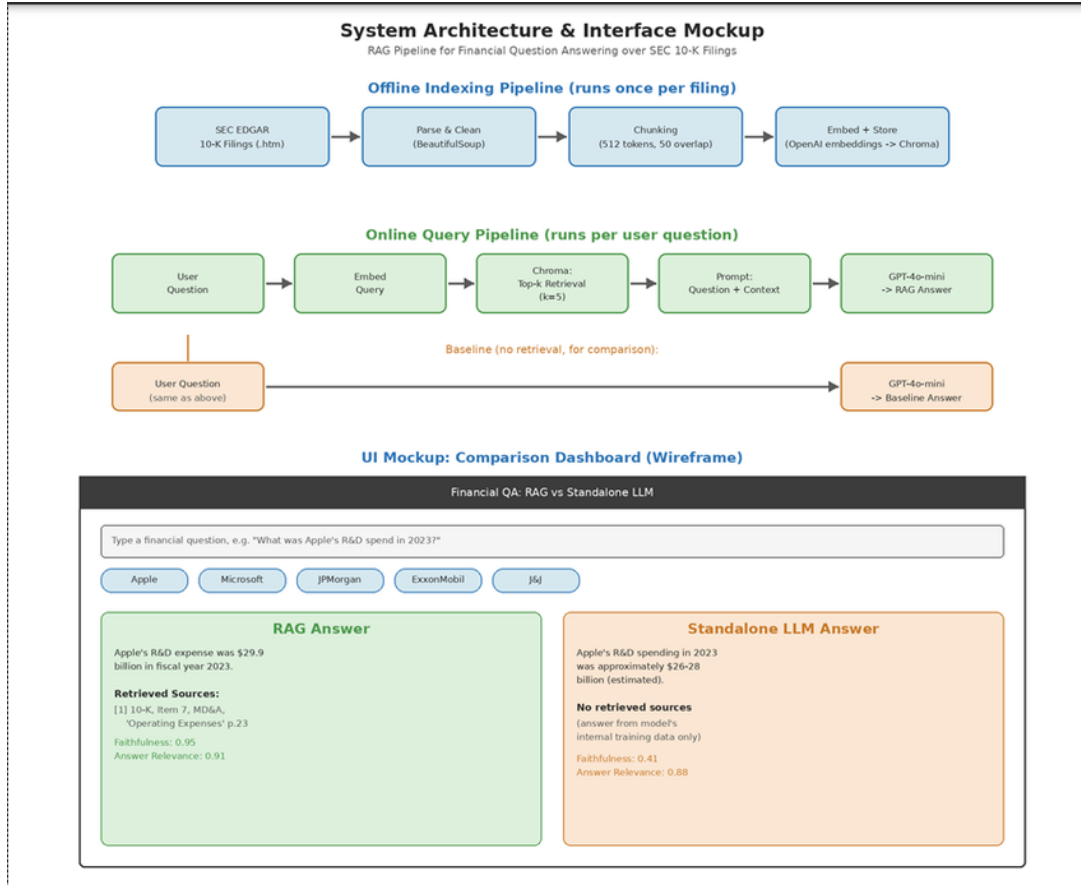


Figure 1: System architecture diagram and UI mockup for the Financial QA comparison interface.

10 Project Ethics and Human Participants

This project does not involve human participants, surveys, or interviews. No personal or sensitive data is collected, processed, or stored. All data used — SEC 10-K filings and the manually constructed evaluation set — is either publicly available regulatory filing data or original content authored from that public data. Confirmed with the project supervisor.

11 BCS Project Criteria

1. **Systematic understanding & critical awareness.** Engages directly with current research (2020–2024) on RAG, RAG evaluation, and financial NLP (Section 4), addressing limitations such as long-context degradation [4].
2. **Comprehensive understanding of applicable techniques.** Applies dense embedding retrieval, vector similarity search, prompt-based generation, and RAGAS evaluation, each selected and justified against documented alternatives in Section 6.
3. **Originality in application of knowledge.** Applying RAG rigorously to FY2023 10-K filings with a purpose-built financial QA dataset and performance breakdown by question type is novel. Combining RAGAS with manual EM/F1 evaluation also goes beyond standard RAG benchmarking.
4. **Handling complex issues, sound judgement, clear communication.** Financial documents present challenges of tabular data, multi-step numerical reasoning, and docu-

ment length. Design decisions (chunk size, retrieval depth) are made under uncertainty and explicitly justified; results are communicated through quantitative metrics and accessible discussion.

5. **Self-direction and originality in problem-solving.** Project scope, dataset construction, evaluation methodology, and technology stack were independently researched and selected, with supervisor consultation used to validate rather than dictate decisions.
6. **Critical self-evaluation.** The dissertation will include an explicit limitations analysis, particularly regarding plain-text RAG on tabular financial data [3], reflecting on which design decisions were most and least effective.

12 Project Plan

The project runs from 1 June to the dissertation deadline of 11 September. Activities overlap iteratively; dissertation writing begins alongside evaluation rather than after it.

Activity	W1	W2	W3	W4	W5–7	W8–11	W12	W13–16
Literature review & research question	X	X						
Architecture & dataset design	X	X	X					
CA1: Specification & Design Proposal				X				
Data collection (5 filings) & cleaning			X	X				
RAG pipeline implementation					X	X		
Baseline LLM pipeline						X		
Evaluation QA dataset (50 Qs)					X	X		
RAGAS + EM/F1 evaluation						X	X	
Results analysis						X	X	
CA2: Presentation video							X	
Dissertation writing						X	X	X
CA3: Dissertation submission								X

Table 2: Project plan (Gantt chart)

13 Risks and Contingency Plans

Risk	Contingency	Likelihood	Impact
OpenAI API costs exceed budget or rate limits are hit	GPT-4o-mini keeps per-token costs low; full evaluation estimated under \$5. A local Ollama-based model (e.g. Llama 3) is kept as a fallback generator.	Low	Medium
SEC EDGAR filing structure varies between companies, breaking the parsing script	Build and test the parsing pipeline on Apple’s filing first; add company-specific handling only where the generic parser fails.	Medium	Medium
RAG performs poorly on numerical / tabular questions	An anticipated limitation documented in the FinQA literature [3]. The evaluation dataset separates question types so poor numerical performance can be reported as a finding rather than project failure.	Medium	Low
Running out of time before the dissertation deadline	Naive RAG was chosen to keep implementation scope realistic. Advanced RAG techniques (re-ranking, query rewriting) are documented as future work, not required deliverables.	Medium	High
Hardware or software failure	All code version-controlled with Git; data and intermediate results (Chroma index, evaluation CSVs) backed up to cloud storage after each work session.	Low	High
Manually constructed evaluation dataset contains errors in ground-truth answers	Each of the 50 QA pairs will be cross-checked against the original filing text at creation time, with the exact source page and section recorded to allow re-verification.	Low	Medium

Table 3: Risk management plan

References

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